

Technical Definitions for Deep Decision Optimization

A Duality Primer for Smoothed SPO+ and Plackett–Luce Compression

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Abstract

This note fixes notation and gives precise definitions for a proposed Deep Decision Optimization (DDO) view of decision-focused learning. The main scientific object is a smoothed version of the SPO+ support-function term. The note is written for a reader who knows optimization but may not have prior exposure to Fenchel conjugacy, support functions, Plackett–Luce ranking models, or mirror-smoothing terminology.

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1 Purpose and one-line thesis

The starting point is the standard predict-then-optimize problem

$$w^*(c) \in \arg \min_{w \in S} c^\top w, \quad (1)$$

where $S \subseteq \mathbb{R}^d$ is a feasible region and $c \in \mathbb{R}^d$ is the true cost vector. A learning model predicts $\hat{c} = f_\theta(x)$ from features x , and the induced decision is $w^*(\hat{c})$.

The SPO+ loss can be written as an affine term plus a support function evaluated at the dual direction $c - 2\hat{c}$:

$$\ell_{\text{SPO}+}(\hat{c}, c) = \sigma_C(c - 2\hat{c}) + 2\hat{c}^\top w^*(c) - c^\top w^*(c), \quad C := \text{conv}(S). \quad (2)$$

Here σ_C is the support function of C .

DDO thesis. Replace the nonsmooth support function σ_C in (2) by a smooth support surrogate $\phi_\tau \approx \sigma_C$, preferably one whose gradient is computable by a compressed moment oracle:

$$\ell_{\text{DDO}}^\tau(\hat{c}, c) := \phi_\tau(c - 2\hat{c}) + 2\hat{c}^\top w^*(c) - c^\top w^*(c). \quad (3)$$

SPO+ is recovered when $\phi_\tau \rightarrow \sigma_C$.

2 Predict-then-optimize, SPO, and SPO+

2.1 The optimization oracle

Definition 1 (Linear optimization oracle). *Let $S \subseteq \mathbb{R}^d$ be nonempty. For any cost vector $a \in \mathbb{R}^d$, define*

$$w^*(a) \in \arg \min_{w \in S} a^\top w. \quad (4)$$

If the minimizer is not unique, $w^(a)$ denotes one selected minimizer. Whenever statements depend on the whole set of minimizers, we write $W^*(a) := \arg \min_{w \in S} a^\top w$.*

Because the objective is linear, optimizing over S and over its convex hull $C = \text{conv}(S)$ gives the same optimal value:

$$\min_{w \in S} a^\top w = \min_{w \in C} a^\top w. \quad (5)$$

This is why convex-analytic objects such as support functions and normal cones naturally enter SPO+ even when S is discrete or mixed-integer.

2.2 SPO loss

Definition 2 (SPO loss). *Given prediction $\hat{c}$ and true cost c , the SPO loss is*

$$\ell_{\text{SPO}}(\hat{c}, c) := c^\top w^*(\hat{c}) - c^\top w^*(c). \quad (6)$$

It measures the true decision regret from solving the optimization problem with $\hat{c}$ instead of c .

The SPO loss is aligned with the final decision objective, but the map $\hat{c} \mapsto w^*(\hat{c})$ is usually piecewise constant, discontinuous at boundaries, and nondifferentiable. This creates weak or unstable gradients for deep learning.

2.3 SPO+ loss

Definition 3 (SPO+ loss). *The SPO+ surrogate is*

$$\ell_{\text{SPO+}}(\hat{c}, c) := \max_{w \in S} \{(c - 2\hat{c})^\top w\} + 2\hat{c}^\top w^*(c) - c^\top w^*(c). \quad (7)$$

It is convex in $\hat{c}$ for fixed c and upper-bounds the SPO loss under the standard SPO setting.

The first term in (7) is the support function of $C = \text{conv}(S)$:

$$\max_{w \in S} (c - 2\hat{c})^\top w = \max_{w \in C} (c - 2\hat{c})^\top w = \sigma_C(c - 2\hat{c}). \quad (8)$$

Thus SPO+ is best understood as a dual/support-function surrogate, not as an ordinary prediction loss.

3 Convex-duality terms

3.1 Indicator function and Fenchel conjugate

Definition 4 (Indicator function). *For a set $C \subseteq \mathbb{R}^d$, the indicator function is*

$$\delta_C(w) = \begin{cases} 0, & w \in C, \\ +\infty, & w \notin C. \end{cases} \quad (9)$$

It converts a constrained optimization problem into an unconstrained extended-valued problem.

Definition 5 (Fenchel conjugate). *For a function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$, its Fenchel conjugate is*

$$f^*(u) := \sup_{w \in \mathbb{R}^d} \{u^\top w - f(w)\}. \quad (10)$$

The vector u lives in the dual space. In finite-dimensional Euclidean space we identify the dual space with $\mathbb{R}^d$ using the dot product.

Interpretation. $f^*(u)$ asks: if a linear function $u^\top w$ tries to support the graph of f , what is the largest vertical gap between $u^\top w$ and $f(w)$? Fenchel conjugacy converts primal constraints and regularizers into dual value functions.

Proposition 1 (Fenchel-Young inequality). *For every $w, u \in \mathbb{R}^d$,*

$$f(w) + f^*(u) \geq u^\top w. \quad (11)$$

Equality holds if and only if $u \in \partial f(w)$, equivalently $w \in \partial f^(u)$, under standard closed convex assumptions.*

3.2 Support function

Definition 6 (Support function). *For a nonempty set $C \subseteq \mathbb{R}^d$, the support function is*

$$\sigma_C(u) := \sup_{w \in C} u^\top w. \quad (12)$$

If C is compact, the supremum is attained.

The support function is exactly the Fenchel conjugate of the indicator function:

$$\sigma_C(u) = \delta_C^*(u). \quad (13)$$

This identity is the key duality bridge in SPO+.

3.3 Subgradient, exposed face, and normal cone

Definition 7 (Subgradient). *For a convex function $g : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$, a vector z is a subgradient of g at u if*

$$g(v) \geq g(u) + z^\top(v - u) \quad \text{for all } v. \quad (14)$$

The set of all subgradients is denoted $\partial g(u)$. If g is differentiable at u , then $\partial g(u) = \{\nabla g(u)\}$.

Definition 8 (Exposed face). *For a closed convex set C and a direction u , the exposed face of C in direction u is*

$$F_C(u) := \arg \max_{w \in C} u^\top w. \quad (15)$$

For compact C , this set is nonempty.

For support functions,

$$\partial \sigma_C(u) = F_C(u). \quad (16)$$

Thus a support-function subgradient is not merely an arbitrary vector; it is a decision point or face of the feasible region selected by a dual direction.

Definition 9 (Normal cone). *For a closed convex set C and point $w \in C$, the normal cone is*

$$N_C(w) := \{u \in \mathbb{R}^d : u^\top(v - w) \leq 0 \text{ for all } v \in C\}. \quad (17)$$

It contains all linear directions u for which w maximizes $u^\top v$ over $v \in C$.

The normal-cone and exposed-face views are equivalent:

$$w \in F_C(u) \iff u \in N_C(w). \quad (18)$$

Geometrically, SPO+ selects a face of C by looking in the dual direction $c - 2\hat{c}$.

4 SPO+ in duality language

Set $u := c - 2\hat{c}$. The SPO+ loss is

$$\ell_{\text{SPO+}}(\hat{c}, c) = \sigma_C(u) + 2\hat{c}^\top w^*(c) - c^\top w^*(c). \quad (19)$$

By (16),

$$\partial_u \sigma_C(u) = F_C(u). \quad (20)$$

By the chain rule for $u = c - 2\hat{c}$,

$$\partial_{\hat{c}} \ell_{\text{SPO+}}(\hat{c}, c) = 2w^*(c) - 2F_C(c - 2\hat{c}). \quad (21)$$

When the exposed face is a singleton $F_C(c - 2\hat{c}) = \{w_{\text{supp}}\}$, the gradient is

$$\nabla_{\hat{c}} \ell_{\text{SPO+}}(\hat{c}, c) = 2(w^*(c) - w_{\text{supp}}). \quad (22)$$

Because

$$\arg \max_{w \in S} (c - 2\hat{c})^\top w = \arg \min_{w \in S} (2\hat{c} - c)^\top w, \quad (23)$$

this support solution can also be computed by the same minimization oracle:

$$w_{\text{supp}} \in W^*(2\hat{c} - c). \quad (24)$$

Preferred terminology. A rigorous phrase is “support-function subgradient” or “exposed-face selector.” The term “adversarial decision” is intuitive but less precise and should be avoided in a formal presentation.

5 Regularized support functions and DDO

5.1 Lower smooth support via regularized Fenchel conjugacy

Definition 10 (Regularizer / mirror potential). *A function $\Omega : C \rightarrow \mathbb{R} \cup \{+\infty\}$ is a regularizer or mirror potential if it is convex and encodes a geometry on C . If Ω is strongly convex, then the regularized maximization below has a unique solution.*

Given a temperature $\tau > 0$, define the regularized support function

$$\sigma_{C,\tau}^\Omega(u) := (\delta_C + \tau\Omega)^*(u) = \sup_{w \in C} \{u^\top w - \tau\Omega(w)\}. \quad (25)$$

When Ω is strongly convex, $\sigma_{C,\tau}^\Omega$ is smooth, and

$$\nabla \sigma_{C,\tau}^\Omega(u) = \arg \max_{w \in C} \{u^\top w - \tau\Omega(w)\} =: w_\tau^\Omega(u). \quad (26)$$

The first-order optimality condition is

$$u - \tau \nabla \Omega(w_\tau^\Omega(u)) \in N_C(w_\tau^\Omega(u)). \quad (27)$$

Compare this with the hard condition $u \in N_C(w)$ for $w \in \partial \sigma_C(u)$. DDO smooths the normal-cone selection rule by adding the mirror-geometry correction $\tau \nabla \Omega(w)$.

Remark 1 (Bias direction). *If $\Omega \geq 0$, then $\sigma_{C,\tau}^\Omega(u) \leq \sigma_C(u)$, so (25) is a lower smooth approximation. It is excellent for differentiability, but it does not automatically preserve the SPO+ upper-bound property. If an upper bound is required, one can add a known correction constant or use the entropy-lifted upper smoothing in the next subsection.*

5.2 Upper smooth support via entropy lifting over finite decisions

Suppose $S = \{w_1, \dots, w_m\}$ is finite. Define the entropy-lifted smoothed support function

$$\phi_{S,\tau}^{\text{ent}}(u) := \tau \log \sum_{j=1}^m \exp\left(\frac{u^\top w_j}{\tau}\right). \quad (28)$$

Equivalently,

$$\phi_{S,\tau}^{\text{ent}}(u) = \max_{q \in \Delta_m} \left\{ \sum_{j=1}^m q_j u^\top w_j + \tau H(q) \right\}, \quad H(q) := - \sum_{j=1}^m q_j \log q_j. \quad (29)$$

The maximizer is

$$q_{\tau,j}(u) = \frac{\exp(u^\top w_j / \tau)}{\sum_{k=1}^m \exp(u^\top w_k / \tau)}. \quad (30)$$

The gradient is the soft support moment

$$\nabla \phi_{S,\tau}^{\text{ent}}(u) = \sum_{j=1}^m q_{\tau,j}(u) w_j =: \bar{w}_\tau(u). \quad (31)$$

The Hessian is a covariance matrix:

$$\nabla^2 \phi_{S,\tau}^{\text{ent}}(u) = \frac{1}{\tau} \text{Cov}_{q_\tau(u)}(W). \quad (32)$$

Proposition 2 (Upper approximation bound). *For finite S ,*

$$\sigma_C(u) \leq \phi_{S,\tau}^{\text{ent}}(u) \leq \sigma_C(u) + \tau \log |S|. \quad (33)$$

Moreover, $\phi_{S,\tau}^{\text{ent}}(u) \rightarrow \sigma_C(u)$ as $\tau \downarrow 0$.

Proof. Let $M = \max_j u^\top w_j = \sigma_C(u)$. Then

$$e^{M/\tau} \leq \sum_j e^{u^\top w_j/\tau} \leq |S| e^{M/\tau}.$$

Multiplying the logarithm by τ gives the result. $\square$

5.3 DDO loss and gradient

Choose a smoothed support surrogate ϕ_τ and define

$$\ell_{\text{DDO}}^\tau(\hat{c}, c) := \phi_\tau(c - 2\hat{c}) + 2\hat{c}^\top w^*(c) - c^\top w^*(c). \quad (34)$$

If $\phi_\tau = \sigma_C$, this is exactly SPO+. If ϕ_τ is differentiable, then

$$\nabla_{\hat{c}} \ell_{\text{DDO}}^\tau(\hat{c}, c) = 2(w^*(c) - \nabla \phi_\tau(c - 2\hat{c})). \quad (35)$$

For the finite entropy-lifted case, this becomes

$$\nabla_{\hat{c}} \ell_{\text{DDO}}^\tau(\hat{c}, c) = 2(w^*(c) - \bar{w}_\tau(c - 2\hat{c})). \quad (36)$$

Thus SPO+ compares $w^*(c)$ with a hard exposed-face selector, whereas DDO compares $w^*(c)$ with a smooth support moment.

6 Plackett–Luce model and score-space compression

6.1 The problem of distributions over rankings

For N items, a full ranking is a permutation

$$\pi = (\pi_1, \dots, \pi_N) \quad (37)$$

of the set $\{1, \dots, N\}$. The number of possible rankings is $N!$. A completely arbitrary distribution over rankings therefore has $N! - 1$ degrees of freedom, which is infeasible even for moderate N .

6.2 Plackett–Luce distribution

Definition 11 (Plackett–Luce model). *Let each item i have a score $s_i \in \mathbb{R}$ and positive weight $\alpha_i = \exp(s_i/\tau)$, where $\tau > 0$ is a temperature. The Plackett–Luce probability of a full ranking π is*

$$\mathbb{P}_{s,\tau}(\pi) = \prod_{t=1}^N \frac{\exp(s_{\pi_t}/\tau)}{\sum_{j=t}^N \exp(s_{\pi_j}/\tau)}. \quad (38)$$

Equivalently, the ranking is generated sequentially: at step t , choose the next item from the remaining set with a softmax probability proportional to $\exp(s_i/\tau)$.

Important properties.

- **Compression:** the model uses N scores rather than $N!$ probabilities.
- **Not universal:** not every distribution over rankings can be represented by Plackett–Luce scores.
- **Shift invariance:** adding the same constant to all scores does not change the distribution.
- **Temperature:** as $\tau \downarrow 0$, rankings concentrate on high-score orderings; as $\tau \rightarrow \infty$, choices become more uniform.

6.3 Geometry behind Plackett–Luce

At each choice step, the softmax map is the gradient of a log-partition function:

$$\Psi_\tau(s; E) = \tau \log \sum_{i \in E} \exp(s_i/\tau), \quad \nabla_{s_i} \Psi_\tau(s; E) = \frac{\exp(s_i/\tau)}{\sum_{j \in E} \exp(s_j/\tau)}. \quad (39)$$

Here E is the set of currently available items. This is the same log-partition/entropy geometry that appears in the entropy-smoothed support function (28).

Why this matters for DDO. Plackett–Luce is not the general DFL solution. Its transferable idea is algorithmic: instead of storing a full distribution over exponentially many decisions, store low-dimensional scores whose mirror/log-partition map gives the needed moments. In a ranking problem, this changes an impossible $O(N!)$ representation into an $O(N)$ score representation.

7 Compressed moment oracles

Definition 12 (Moment oracle). *A moment oracle for a smoothed support surrogate ϕ_τ is a procedure that returns*

$$M_\tau(u) := \nabla \phi_\tau(u) \quad (40)$$

without enumerating all feasible decisions. In DDO, $M_\tau(c - 2\hat{c})$ replaces the hard support-function subgradient $\partial\sigma_C(c - 2\hat{c})$.

Examples:

- **Finite enumerated set:** compute $\bar{w}_\tau = \sum_j q_j w_j$ exactly.
- **Ranking:** compute PL/softmax moments using N item scores rather than $N!$ ranking probabilities.
- **Shortest path with structure:** compute soft path marginals using dynamic programming when available.
- **Black-box solver:** approximate the moment using a candidate pool, perturbed solver calls, or cached feasible decisions.

The compressed moment oracle is the algorithmic part of DDO. The smoothed support formula is the duality part.

8 Technical terms table

Term	Symbol	Meaning and role in DDO
Feasible set	S	Set of allowed decisions. It can be discrete, continuous, or mixed-integer.
Convex hull	$C = \text{conv}(S)$	Smallest convex set containing S . Linear optimization has the same optimal value over S and C .
Optimization oracle	$w^*(a)$	A decision minimizing $a^\top w$ over S . SPO and SPO+ call this oracle at different cost vectors.
Prediction	$\hat{c} = f_\theta(x)$	Model output used as the cost vector in the downstream optimization problem.
SPO loss	ℓ_{SPO}	True decision regret $c^\top w^*(\hat{c}) - c^\top w^*(c)$.
SPO+ loss	$\ell_{\text{SPO+}}$	Convex surrogate using $\sigma_C(c - 2\hat{c})$ plus an affine correction.
Indicator function	δ_C	Extended-valued function equal to 0 on C and $+\infty$ outside C .
Fenchel conjugate	f^*	Dual function $f^*(u) = \sup_w \{u^\top w - f(w)\}$. Turns indicators and regularizers into support-like functions.
Support function	σ_C	$\sigma_C(u) = \max_{w \in C} u^\top w = \delta_C^*(u)$. This is the central dual object in SPO+.
Dual direction	$u = c - 2\hat{c}$	Direction at which SPO+ evaluates the support function.
Subgradient	$\partial g(u)$	Set of valid first-order slopes for a convex nonsmooth function. For support functions, subgradients are exposed-face points.
Exposed face	$F_C(u)$	$\arg \max_{w \in C} u^\top w$. SPO+ selects this face through $\partial \sigma_C(u)$.
Normal cone	$N_C(w)$	Directions u for which w maximizes $u^\top v$ over C . Equivalent condition: $w \in F_C(u) \Leftrightarrow u \in N_C(w)$.
Hard support oracle	$\partial \sigma_C(u)$	The nonsmooth SPO+ comparison object. It can jump between vertices or faces as u changes.
Regularizer	Ω	Convex function encoding geometry. Strong convexity makes the smoothed support differentiable.
Mirror potential	Ω	Same object as a regularizer, emphasized as defining a non-Euclidean geometry for updates.
Bregman divergence	$D_\Omega(x, y)$	Geometry-induced distance: $\Omega(x) - \Omega(y) - \nabla \Omega(y)^\top (x - y)$. Used in mirror descent.
Temperature	τ	Smoothing parameter. Smaller τ is closer to hard SPO+; larger τ gives smoother gradients.
Regularized support	$\sigma_{C, \tau}^\Omega$	Smooth conjugate $(\delta_C + \tau \Omega)^*$; gives a differentiable support selector.
Entropy	$H(q)$	$-\sum_j q_j \log q_j$. Encourages spread-out distributions and yields log-sum-exp smoothing.
Log-partition	$\tau \log \sum_j e^{a_j/\tau}$	Smooth maximum of scores a_j . Its gradient is softmax.
Softmax	$q_j \propto e^{a_j/\tau}$	Probability distribution over alternatives induced by scores.
Soft support moment	$\bar{w}_\tau$	Expected decision $\sum_j q_j w_j$. This replaces a hard support-function subgradient in DDO.
Plackett–Luce model	$\mathbb{P}_{s, \tau}(\pi)$	Ranking distribution generated by repeated softmax choices. It compresses a ranking distribution from $N!$ probabilities to N scores.

Term	Symbol	Meaning and role in DDO
Score-space compression	$s \in \mathbb{R}^N$	Store scores, not all ranking probabilities. This is the algorithmic lesson borrowed from RIPLM/PL.
Moment oracle	$M_\tau(u)$	Procedure returning $\nabla\phi_\tau(u)$ efficiently. It is what makes smoothed SPO+ computationally practical.
Candidate set	$\mathcal{C} \subseteq S$	Small set of feasible decisions used to approximate the full soft support when exact computation is impossible.
Annealing	$\tau \downarrow 0$ during training	Start with smooth gradients, then move toward the original SPO+ objective.
Upper smoothing	$\phi_\tau \geq \sigma_C$	Preserves the SPO+ upper-bound spirit, usually with a controllable additive bias.
Lower smoothing	$\phi_\tau \leq \sigma_C$	Useful for differentiability but may need a correction constant if an upper bound is required.

9 What to say precisely in a presentation

A mathematically careful description is:

SPO+ is an affine correction plus the support function $\sigma_C = \delta_C^*$ evaluated at the dual direction $c - 2\hat{c}$. Its subgradient is an exposed-face selector $\partial\sigma_C(c - 2\hat{c})$. DDO replaces this nonsmooth support-function subgradient with a smooth support moment $\nabla\phi_\tau(c - 2\hat{c})$. When $\phi_\tau \rightarrow \sigma_C$, DDO recovers SPO+. The algorithmic question is whether the smooth moment can be computed without enumerating the decision space. Plackett–Luce gives one example of such compression for rankings: it represents a distribution over $N!$ rankings using N scores.

10 References for context

References

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